Round 10
The young-source cost bound

One focused assessment · n = 1024, L = 16 · 24 September 2026

Answer: no justified 55 u construction

I do not have a different representation that meets both ≤55 u and ≤2% D21 error. A precise negative result is possible for the dense-leg contraction model in your question: accepting your exhaustive contraction-order result, its product calls alone cost at least about **98.43 u**, even allowing the final layer to compute D3 only. This is a conditional bound for that computational model, **not a proof against every approximate representation**.

Derived, conditional claim. Fix the carried legs, birth coefficients and source identities. Permit the specified dense products, elementwise operations, associativity, intermediate reuse and batching, with the supplied level-5 square-product primitive. Assume, as your search reports, that an interior source needs two transport products and two fused right-factor contractions. Under these assumptions the 55 u target is impossible. A different nonlinear approximation is outside this claim.

The state and the unavoidable calls in this model

At layer ℓ let $m_\ell=\min(4,\ell-1)$ young sources be active. Their persistent state is $2m_\ell$ dense $n\times n$ legs and m_ℓ birth-weight vectors: at four sources, $8n^2+4n$ float32 numbers, or about 32.016 MiB. Count the newborn identity shortcut P=W, which you already use. Birth-matrix construction, thin feedback, joins and the old tier remain outside this young-only count.

$$g(n)=n^2(2n-1), \qquad N_{\text{transport}}=2m_\ell-1, \quad N_{\text{slice}}=2m_\ell.$$

Layer(s)	Young sources	Square products per layer
2	1	1 transport + 2 slice = 3
3	2	3 transport + 4 slice = 7
4	3	5 transport + 6 slice = 11
5-16	4	7 transport + 8 slice = 15

$$\sum_{\ell=2}^{16} m_\ell = 54, \qquad N_{\text{full}} = 3+7+11+12(15) = 201.$$

If layer 16 needs only D3, delete its eight slice products: **N=193**. This deliberately gives the implementation the cheaper terminal branch. If your deliverable literally requires full D21 there too, use 201. With S_5 the actual metered FLOPs of one level-5 square product, the product-only bound is $193\,S_5/2^{31}$ u. At $S_5/2^{31}\approx0.51$ this is 98.43 u, versus 102.51 u for all full slices.

Even before elementwise arithmetic, reaching 55 u with 193 calls would require ≤0.28497 u per call, or eliminating approximately 45% of the current calls. Batching does not supply that elimination under the stipulated counting rule.

An exact ledger, and the boundary of the proof

Derived: explicit arithmetic schedule. Form M=W diag(Φ) once (n^2 multiplies); transport old young legs with M and the newborn A with W. For each transported source form P^2, PA, their two column-weighted versions, and multiply the second by 2: $5n^2$ operations. Form the two n×n products, accumulate all 2m outputs, and divide the final sum by 3. Extracting its diagonal needs no arithmetic. This particular schedule has the exact cost

$$F_\ell = (4m_\ell - 1)S_5 + (7m_\ell + 1)n^2.$$

Here S_5 includes all additions internal to that square-product implementation. Do not add the same Strassen overhead twice. The formula specifies a reproducible schedule, not an assertion that it is the shipped meter's exact decomposition; cheaper elementwise scheduling only reduces the small n^2 term.

For the terminal D3-only branch, use $d3_i=\Sigma_{sj} w_{sj} P_{s,ij}^2 A_{s,ij}$. Its exact cost in this schedule is $(2m-1)S_5+(4m+1)n^2-n$. Therefore:

Requested readout	Exact network FLOPs	Nominal u if $S_5/2^{31}=0.51$
Full D21 at all 15 transitions	201 S_5 + 393 n^2	102.701895
Final transition D3 only	193 S_5 + 381 n^2 − n	98.616035

For classical square products S_5 is replaced by g=2,146,435,072 FLOPs: the corresponding exact totals are 431,845,539,840 and 414,661,475,328 FLOPs. Your rounded 0.51 u does not specify an exact integer Strassen bill; substitute its measured S_5 in the formulas. Extra source/birth operations can only increase this schedule's bill.

Why the suggested regroupings do not evade this certificate. Transporting $[P_1,A_1,\ldots,P_m,A_m]$ has shape (n,n)×(n,2mn), costing 2m g classically before the newborn shortcut. Combining the slice products has shape (n,2mn)×(2mn,n), costing $n^2(4mn-1)$. Equivalently, 2m square calls plus $(2m-1)n^2$ accumulation operations. Calling either a "single product" does not make it a single square-product unit.

Shared propagator factors do not close their squares:

$$(UV)_{ij}^2 = \sum_a U_{ia}^2 V_{aj}^2 + 2\sum_{a<b} U_{ia}U_{ib}V_{aj}V_{bj}.$$

Keeping those cross terms explicitly requires a contraction of shapes (n,q)×(q,n), q=n(n−1)/2=523,776. That contraction alone costs $n^2(2q-1)/2^{31}$=511.499512 u at n=1024, before feature formation. The original dense-leg route is much cheaper. A closed age-2 expression merely chooses how to evaluate these same factors unless it introduces a new approximation.

Accuracy boundary. These identities preserve D21 exactly in real arithmetic, but exceed the cost target. Neither flat spectra nor a contraction-order search proves a lower bound for *every* 2%-accurate algorithm. The allowed error corresponds to $4.2\times10^{-6}(0.02)^2=1.68\times10^{-9}$ extra final MSE under your measured law. No unmeasured replacement is claimed to satisfy it.

One falsifier: test the stated transport model without an inverse

Derived consistency condition. There is a conflict between two stated premises. If a particular source obeys $P_+=MP$ and $A_+=MA$ with the same M, then any exact relation $A=PR$ is preserved: $A_+=P_+R$. If P and M are invertible, $P_+^{-1}A_+=P^{-1}A$ exactly. Therefore a 100-970% change in numerically computed $P^{-1}A$ does not, by itself, establish a changing right factor. An ill-conditioned inverse, source misalignment, a change of coordinates, or an unrecorded update can produce that observation.

I still charge A's transport. Even if this invariant holds, evaluating $A_+=P_+R$ is another dense multiplication, so it does not yield the requested saving. The issue matters because a no-go statement must specify the actual computation it covers.

Measure. Select one matched birth at age 2→3, away from the join, and capture both legs immediately before and after the same transport. Preserve birth identity and coordinate conventions. Use float64 products on the captured float32 values; compute

$$\rho_P = \frac{\|P_+ - MP\|_F}{\|P_+\|_F}, \qquad \rho_A = \frac{\|A_+ - MA\|_F}{\|A_+\|_F}.$$

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M  = W.astype(np.float64) * phi[None, :]
P0 = P_before.astype(np.float64)
A0 = A_before.astype(np.float64)
P1 = P_after.astype(np.float64)
A1 = A_after.astype(np.float64)
RP, RA = P1 - M @ P0, A1 - M @ A0
rhoP = np.sqrt(np.sum(RP * RP) / np.sum(P1 * P1))
rhoA = np.sqrt(np.sum(RA * RA) / np.sum(A1 * A1))
```

Interpretation. If ρ_A substantially exceeds the transport's forward-roundoff scale while ρ_P does not, the stipulated same-left-multiplication model is false for A at that capture boundary; an innovation or unmatched convention must be included, and the certificate above does not describe that graph. If both residuals are at roundoff scale, the inverse-based drift does not refute exact common transport. Establish the roundoff scale by replaying the same operation, rather than using a universal fixed tolerance.

Exact diagnostic cost. For the shown algorithm, treating sqrt/division as one arithmetic operation each and charging float64 twice, the arithmetic is $4g+22n^2$ = **8,608,808,960 FLOPs = 4.0087890625 u**. Casts and copies are excluded under the stated arithmetic model. It uses two classical 1024-square products and $O(n^2)$ storage; one stored transition should take well under an hour on a CPU that runs the benchmark.

This is a falsifier of the *model underlying the conditional bound*, not a test of a secretly proposed 55 u algorithm. No such algorithm has been supplied here. Your leaderboard result demonstrates a cheaper successful estimator, but does not establish that the gain is achievable by changing only ages 1-4 while freezing the rest of V29.

Conclusion

The requested target remains open. The supplied facts support exclusion of the specified exact dense-leg regroupings. They do not support a universal impossibility proof for different approximate representations, and I cannot honestly provide a representation with both the requested cost and an evidence-based ≤2% error claim.

Evidence scope: network measurements and the exhaustive contraction-order result are supplied by the user. Counts and identities above are derived here. No stored network captures were supplied for this response, so the falsifier has not been run on the network. No additional literature-based accuracy claim is made.